Limitations

This work only scratches the surface of spatial reasoning of LLMs. Both *mental images* and visual state tracking rely on the emergent ability of advanced LLMs. Therefore, it might cause performance deterioration in less advanced language models or more challenging tasks. Besides, due to the limited data exposure and a lack of explicit instruction tuning, visual state tracking of current LLMs are sensitive to prompts. For example, when explicitly prompted with "use ascii-art", the tracking rate will significantly increase thereby boosting performance, while removing "reasoning" from the **VoT** prompt will cause a decrease of tracking rate. Moreover, the *mental images* tested in our work are limited to 2D grid. To strength the mind's eye of LLMs, more diverse and complicated representation should be explored in the future, such as complex geometric shapes and even 3D semantics shown in Figure 11 in appendix.

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